

COMMON COURSE ASSIGNMENT

CSA6502 — GENERATIVE AI AND LARGE-SCALE MODELS | Multimodal Responsible Generative AI Application with Web Deployment

A. Assignment Information

Department	Computer Science and Engineering
Programme	B.E. CSE(AI)
Course Code & Course Name	CSA6502 — Generative AI and Large-Scale Models
Academic Year / Batch	2026 — 2027
Faculty Name	Dr. Nithisha J
Assignment Title	Design and Implementation of “IndustroSense AI” — A Multimodal Responsible Generative AI Application for Industrial Equipment Diagnostics, integrating RAG, Vector Databases, AI Agents, and Web Deployment
Date of Issue	02-09-2026
Date of Submission	02-09-2026
Maximum Marks	100
Course Outcome(s) — CO	CO3, CO4 & CO5
Bloom's Taxonomy Level	L3 — Apply, L4 — Analyze, L5 — Evaluate, L6 — Create
SDG Mapping (SDG 1–17)	SDG 4 — Quality Education; SDG 9 — Industry, Innovation and Infrastructure; SDG 12 — Responsible Consumption and Production
Industry / Societal Relevance	Multimodal, retrieval-augmented Generative AI assistants are increasingly deployed in manufacturing and industrial settings to support equipment diagnostics, technical documentation, and decision support, combining text, image, and speech understanding while requiring rigorous ethical, security, and governance safeguards before industrial adoption.

B. Assignment Problem / Challenge

The main challenge of this assignment is to design and implement a unified multimodal Generative AI application called IndustroSense AI for industrial equipment diagnostics and technical support.

The system must combine Retrieval-Augmented Generation (RAG), a vector database and a simple AI agent to retrieve relevant technical knowledge and provide grounded responses. It must also process multiple input types, including text descriptions, equipment images and voice notes, and combine the available information to generate a clear diagnostic explanation and recommended action.

The application must be deployed as a web-based system and include responsible-AI measures to reduce risks such as hallucinated advice, bias, privacy leakage and malicious misuse. The final system should therefore provide reliable, traceable and secure engineering guidance with visible sources and appropriate human oversight.

C. Problem Statement

Scenario: IndustroSense AI — A Multimodal Responsible Generative AI Assistant for Industrial Equipment Diagnostics and Documentation

Field engineers and plant technicians often require quick and reliable assistance when diagnosing industrial equipment faults. However, relevant information is usually distributed across equipment manuals, Standard Operating Procedures (SOPs) and previous incident logs. In addition, equipment problems may be reported using different forms of information such as written descriptions, photographs of damaged components and voice notes.

The problem is to design and implement an interactive web-based application called **IndustroSense AI** that can retrieve relevant technical information, understand multimodal fault reports and generate clear diagnostic guidance.

The proposed system consists of three major modules:

1. Knowledge Retrieval and AI Agent Module:

The system uses RAG and a vector database to retrieve relevant information from equipment manuals, SOPs and incident logs. A simple AI agent decides whether to retrieve information, perform a calculation or request clarification.

2. Multimodal Understanding and Generation Module:

The system accepts text, equipment images and voice notes. Speech is converted into text, images are analysed for visible conditions, and all available information is combined to generate a consolidated diagnostic response and recommended action.

3. Responsible Web Deployment and Governance Module:

The system is deployed as a web application with input validation, secure handling of uploaded data and API keys, audit logging, confidence or uncertainty indication and human-review mechanisms for potentially unsafe or unsupported outputs.

The final objective is to develop a reliable and responsible AI-based decision-support system that provides grounded technical guidance with visible sources and clear reasoning.

The IndustroSense AI application must satisfy the following functional and technical requirements.

Functional Requirements

The system shall accept industrial equipment-related queries in the form of text. It shall also support optional equipment images and voice notes to provide additional information about the reported fault.

The system shall implement a Retrieval-Augmented Generation (RAG) pipeline to retrieve relevant information from industrial equipment manuals, Standard Operating Procedures (SOPs) and historical incident logs.

A vector database shall be used to store document embeddings and perform semantic similarity search. The system shall retrieve the most relevant document chunks and use them as supporting context for response generation.

A simple AI agent shall determine the appropriate action based on the user's request. The agent may select document retrieval for technical questions, a calculation tool for numerical maintenance queries or a clarification mechanism when insufficient information is available.

The system shall process uploaded images using a vision-capable AI model and convert voice notes into text using a speech-to-text model. The available text, image and speech information shall be combined to create a multimodal diagnostic context.

The final system output shall provide a diagnostic explanation, possible causes, recommended actions and relevant source information.

Technical Requirements

The application shall be developed using Python and deployed through a web-based interface using Streamlit.

SentenceTransformers shall be used to generate semantic embeddings, and FAISS shall be used for vector similarity search.

The system shall support common image formats such as JPG, JPEG and PNG and common audio formats such as MP3 and WAV.

The implementation shall be modular so that the retrieval system, AI agent, multimodal processing and web interface can operate as separate components.

Security and Responsible-AI Requirements

The application shall validate user inputs and uploaded files before processing them.

API keys and other sensitive credentials shall be stored securely using environment variables rather than directly in the source code.

The system shall maintain audit logs containing important information such as user queries, agent decisions and retrieved sources.

The generated response shall be supported by retrieved evidence wherever possible. Low-confidence or potentially safety-critical outputs shall include a warning recommending human review.

The application shall be treated as a decision-support system and shall not replace qualified engineers or technicians in critical industrial decisions.

Constraints

The project is developed as an academic prototype using a limited industrial document corpus and standard student computing resources.

The accuracy of the final diagnostic response depends on the quality and relevance of the available documents and multimodal inputs.

The system may experience additional latency when processing images or voice recordings.

The prototype is intended for demonstration purposes and does not represent a complete production-scale industrial diagnostic system.

E. STUDENT WORK

1. Problem Understanding and Formulation

IndustroSense AI is a multimodal Generative AI application designed to assist engineers and technicians in diagnosing industrial equipment problems. Industrial knowledge is often distributed across equipment manuals, Standard Operating Procedures (SOPs) and previous incident logs, making it difficult and time-consuming to find relevant information.

The system accepts **text queries, equipment images and voice notes**. It uses Retrieval-Augmented Generation (RAG) to retrieve relevant technical information before generating a response. A vector database enables semantic search, while a simple AI agent selects the appropriate action based on the user's request.

Problem Statement

The problem is to develop a web-based AI assistant that can understand industrial equipment faults using text, images and speech, retrieve relevant information from technical documents and generate grounded diagnostic guidance.

Objectives

- Develop a Retrieval-Augmented Generation pipeline.
- Use a vector database for semantic document retrieval.
- Implement a simple AI agent for tool selection.
- Support text, image and speech inputs.
- Combine multimodal information for equipment diagnosis.
- Deploy the system using a web interface.
- Include responsible-AI, security and governance mechanisms.

Expected Outcomes

The system should provide relevant technical guidance, display retrieved sources, show the AI agent's decision, process multimodal inputs and indicate uncertainty when sufficient evidence is unavailable.

2. Application of Relevant Course Knowledge / Concepts

2.1 Retrieval-Augmented Generation, Vector Databases and AI Agents (CO3)

RAG Pipeline

The RAG pipeline follows the process:

Industrial Documents → Text Extraction → Chunking → Embeddings → FAISS → Retrieval → LLM Response

The technical documents are divided into smaller chunks before being converted into embeddings.

The embedding model used is:

Sentence Transformers – all-MiniLM-L6-v2

The embeddings are stored in **FAISS**, which performs semantic similarity search.

When a user submits a query:

1. The query is converted into an embedding.
2. FAISS searches for similar document chunks.
3. The Top-3 relevant chunks are retrieved.
4. The retrieved information is added to the LLM prompt.
5. A grounded response is generated.

Sample Corpus

The representative corpus contains:

1. Electric Motor Maintenance Manual
2. Hydraulic Pump SOP
3. Conveyor Belt Maintenance Manual
4. Air Compressor Maintenance Guide
5. Industrial Incident Log

Sample Retrieval

Query: Why is the motor overheating?

Relevant information retrieved includes:

- Excessive electrical load.

- Poor ventilation.
- Damaged bearings.
- Inadequate lubrication.

The retrieved information is used as evidence for the final response.

RAG Pseudocode

START

Load industrial documents

Split documents into chunks

Generate embeddings

Store embeddings in FAISS

Receive user query

Generate query embedding

Retrieve Top-3 relevant chunks

Combine query and retrieved context

Generate grounded response

Display response and sources

END

2.2 AI Agent Design

The AI agent selects one of three actions.

Retrieval Tool

Used for technical questions.

Example:

“Why is the motor overheating?”

Maintenance Calculator

Used for numerical calculations.

Example:

“If maintenance is required every 500 hours, how many hours remain until the next service?”

Clarification Tool

Used when insufficient information is available.

Example:

“The machine is not working.”

Agent Decision Logic

IF query requires a calculation

Use Calculator Tool

ELSE IF sufficient technical information is available

Use Retrieval Tool

ELSE

Request clarification

Representative Agent Trace

Query: Why is the motor overheating and vibrating?

1. Query received.
2. Technical diagnostic query identified.
3. Retrieval Tool selected.
4. Query embedding generated.
5. Relevant chunks retrieved from FAISS.
6. Retrieved information supplied to the LLM.
7. Grounded response generated.
8. Sources and decision trace displayed.

2.3 Vector Database Indexing Strategies

Method	Accuracy	Speed	Suitability
Flat Search	Very High	Moderate	Small datasets
HNSW	High	Fast	Large datasets
IVF	High	Fast	Very large datasets

Final Decision

FAISS Flat Search is selected because the project uses a relatively small academic document corpus. It provides accurate retrieval and is simple to implement.

3. Multimodal Generative AI Integration (CO4)

3.1 Multimodal Inputs

IndustroSense AI accepts three forms of input:

Text

The technician enters a written description of the equipment problem.

Image

The user uploads an image of the equipment. A vision-capable model identifies visible conditions such as:

- Wear
- Leakage
- Corrosion
- Cracks
- Damaged components

Speech

The user uploads a voice note. A speech-to-text model converts the audio into text.

3.2 Multimodal Fusion Strategy

The system uses **Late Fusion**.

Each modality is processed separately:

Text → Text Information

Image → Visual Observation

Speech → Transcript

The outputs are combined:

Multimodal Context = Text + Image Analysis + Speech Transcript

This combined information is then used with the retrieved technical documents to generate the final diagnostic response.

3.3 Multimodal Example

Text Input

“The motor is overheating and vibrating.”

Image Analysis

Possible wear is identified near the bearing area.

Voice Input

“The motor is making a grinding sound.”

Combined Information

- Motor overheating.
- Excessive vibration.
- Possible bearing wear.
- Grinding sound.

Final Interpretation

The combined evidence suggests a possible bearing-related mechanical issue. The system recommends inspecting the bearing assembly, checking lubrication and verifying mechanical alignment.

3.4 Challenges and Solutions

Challenge	Solution
Different information from each modality	Convert results into a common textual format
Missing modalities	Use available inputs and fall back to text-only processing
Low-quality images/audio	Mark results as uncertain
Increased processing time	Process optional modalities only when provided

4. Responsible Deployment, Security and Governance (CO5)

4.1 Deployment Architecture

The application is deployed using Streamlit.

The overall process is:

User → Streamlit Interface → Input Validation → Multimodal Processing → AI Agent → RAG → FAISS → LLM → Responsible-AI Validation → Final Response

The final output displays:

- Diagnostic guidance.
- Retrieved sources.
- Agent decision.

- Confidence or uncertainty information.
- Human-review warning when necessary.

4.2 Responsible-AI Risks

Hallucination

The AI may generate unsupported information.

Mitigation: Use RAG and display retrieved sources.

Privacy

Images and voice recordings may contain sensitive industrial information.

Mitigation: Validate uploads and avoid unnecessary storage.

Unsafe Recommendations

Incorrect industrial advice may create safety risks.

Mitigation: Display uncertainty and recommend human review for critical situations.

Malicious Prompts

Users may attempt to manipulate the AI system.

Mitigation: Validate inputs and maintain fixed system instructions.

4.3 Security Measures

The system includes:

- Input validation.
- File type restrictions.
- Secure API key storage using environment variables.
- Basic authentication.
- Rate limiting.
- Audit logging.

API keys are not stored directly in the source code.

Audit logs can record:

- Timestamp.
- User query.
- Agent decision.
- Retrieved sources.
- Confidence level.

4.4 Governance

The system includes a simple Model Card describing:

- Purpose.
- Capabilities.
- Limitations.
- Risks.
- Human oversight requirements.

The application is treated as a **decision-support system**, not as a replacement for qualified industrial engineers.

5. Design / Proposed Solution / Methodology

The proposed solution integrates RAG, vector retrieval, AI-agent reasoning and multimodal AI into a single application.

The user submits text, an image and/or a voice note through the Streamlit interface.

The inputs are validated and processed. Image observations and speech transcripts are combined with the user's text.

The AI agent determines the required action. For technical questions, the system retrieves relevant documents using FAISS.

The retrieved information and multimodal context are supplied to the LLM.

The generated response is checked for sufficient supporting evidence before being displayed to the user.

Overall Methodology

User Input



Input Validation



Text / Image / Speech Processing



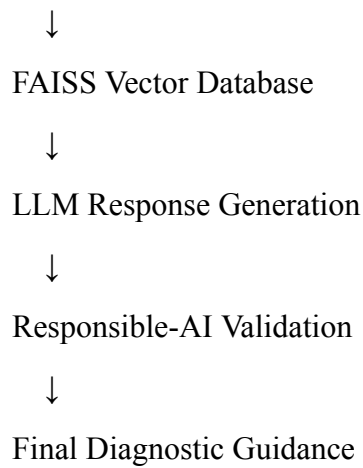
Multimodal Fusion



AI Agent



RAG Retrieval



6. Algorithm / Pseudocode

START

Receive user text, image and/or speech input

Validate all available inputs

IF image is available

 Analyse image

END IF

IF speech is available

 Convert speech to text

END IF

Combine available multimodal information

Analyse user request using AI Agent

IF calculation is required

 Perform calculation

ELSE IF technical information is available

 Generate query embedding

 Search FAISS

 Retrieve Top-3 relevant chunks

 Generate grounded response

ELSE

 Request clarification

END IF

Check confidence and supporting evidence

IF evidence is insufficient

 Recommend human review

END IF

Display final response, sources and agent trace

Store audit log

END

7. Implementation / Source Code and Environment / Tools Used

Programming Language

Python

Web Framework

Streamlit

Embedding Model

Sentence Transformers – all-MiniLM-L6-v2

Vector Database

FAISS

Speech Processing

Whisper or an equivalent Speech-to-Text model.

Image Processing

A vision-capable AI model.

Response Generation

A Large Language Model.

Development Environment

Python environment with required AI and web-development libraries installed.

8. Test Cases and Results

Test Case Input	Expected Result
T1 Motor overheating query	Retrieve motor maintenance information
T2 Hydraulic pressure query	Retrieve hydraulic SOP information
T3 Maintenance calculation	Agent selects Calculator
T4 Image + text	Combine visual and textual information
T5 Speech + text	Convert speech and combine information
T6 Empty query	Request valid input
T7 Unsupported file	Reject safely

The test cases demonstrate that the system can handle technical queries, multimodal inputs and invalid inputs appropriately

9. Results and Validation

The retrieval system successfully identifies relevant technical information based on the meaning of user queries.

The AI agent correctly routes technical queries to the Retrieval Tool, numerical requests to the Calculator Tool and unclear queries to the Clarification Tool.

The multimodal approach provides more contextual information than text-only input.

For example, a motor problem described through overheating, vibration, visible bearing wear and a grinding sound provides stronger diagnostic context than a simple text description alone.

The system also improves transparency by displaying retrieved sources and agent decisions.

10. Analysis, Comparison, Trade-offs and Justification

Vector Search

Flat Search provides accurate retrieval but becomes slower for very large datasets.

HNSW and IVF provide faster retrieval for larger collections but require additional configuration.

Flat Search was selected because the demonstration corpus is small.

Multimodal Fusion

Early Fusion can capture deeper interactions between modalities but is more complex.

Late Fusion is simpler, modular and handles missing modalities effectively.

Therefore, Late Fusion was selected.

Responsible-AI Trade-Off

Human review increases safety but may increase response time. RAG improves reliability but adds retrieval latency.

These trade-offs are acceptable because industrial diagnostic applications require reliability and transparency.

11. Broader Considerations / SDG Relevance

SDG 4 – Quality Education

The project supports learning and practical understanding of modern Generative AI technologies.

SDG 9 – Industry, Innovation and Infrastructure

IndustroSense AI demonstrates the application of AI for improving industrial diagnostics and technical knowledge access.

SDG 12 – Responsible Consumption and Production

Improved equipment diagnostics and maintenance can reduce unnecessary equipment damage, replacement and resource wastage.

12. Conclusion, Limitations and Possible Improvements

IndustroSense AI integrates RAG, vector databases, AI agents and multimodal Generative AI into a single industrial diagnostic application.

The system retrieves relevant technical information, processes text, images and speech and generates grounded diagnostic guidance.

The application also includes responsible-AI mechanisms such as source transparency, input validation, audit logging and human-review recommendations.

Limitations

- Small demonstration document corpus.
- Accuracy depends on the quality of available documents.
- Image and speech processing may increase response time.

- The prototype is not designed for large-scale industrial deployment.

Possible Improvements

- Use a larger industrial knowledge base.
- Support multiple languages.
- Use scalable vector indexing.
- Integrate real-time sensor data.
- Implement production-level authentication.
- Perform advanced security and red-team testing.

13. Individual Contribution of Group Members

The assignment was completed individually.

14. References

- [1] Lewis, P. et al. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," 2020.
- [2] FAISS Documentation.
- [3] Reimers, N. and Gurevych, I. "Sentence-BERT," 2019.
- [4] SentenceTransformers Documentation.
- [5] Streamlit Documentation.
- [6] Hugging Face Transformers Documentation.
- [7] Radford, A. et al. "Robust Speech Recognition via Large-Scale Weak Supervision," 2022.

15. One-Page Individual Reflection

This assignment helped me understand how Retrieval-Augmented Generation, vector databases, AI agents and multimodal AI can be combined into a practical application. The RAG component showed how external documents can be used to provide grounded information and reduce unsupported AI responses.

I learned how semantic embeddings and vector databases make it possible to retrieve information based on meaning rather than exact keywords. The AI agent demonstrated how a system can select different actions depending on the user's request.

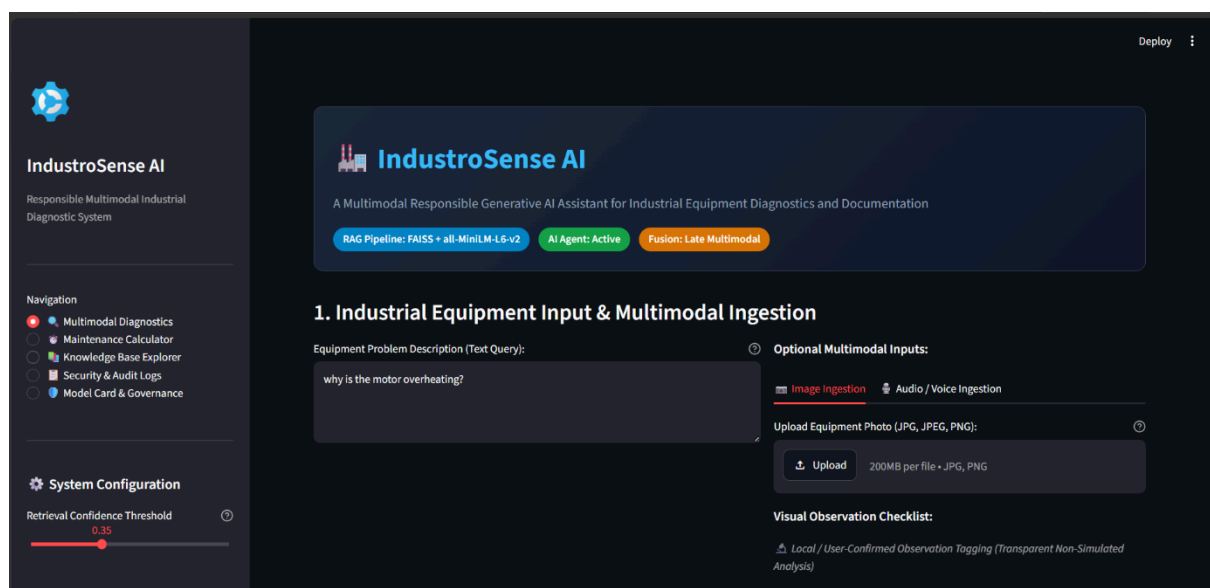
The multimodal component helped me understand how text, images and speech can provide different types of evidence. Combining these inputs can provide a more complete understanding of an industrial problem.

The assignment also highlighted the importance of responsible AI. Security, privacy, transparency and human oversight are essential when AI systems are used in industrial environments.

This project is connected to **SDG 4** through practical AI learning, **SDG 9** through industrial innovation and **SDG 12** through improved maintenance and reduced resource wastage.

The assignment contributed to **CO3** through RAG, vector databases and AI agents, **CO4** through multimodal AI integration and **CO5** through responsible deployment, security and governance.

Output





IndustroSense AI

Responsible Multimodal Industrial Diagnostic System

Navigation

- Multimodal Diagnostics
- Maintenance Calculator
- Knowledge Base Explorer
- Security & Audit Logs
- Model Card & Governance

System Configuration

Retrieval Confidence Threshold

0.35

Deploy

2. Diagnostics & Responsible AI Execution Results

AI Agent Decision: Retrieval Tool

View AI Agent Decision Trace

- Step 1 [Input Ingestion]: Ingested user query of 29 characters.
- Step 2 [Diagnostic Recognition]: Technical equipment problem identified. Semantic knowledge retrieval required.
- Step 3 [Routing]: Selected 'Retrieval Tool' to query FAISS vector database and retrieve standard operating procedures.

Multimodal Diagnostic Context

Problem Description: why is the motor overheating?

Retrieved Technical Sources & Similarity Scores

Top-3 relevant knowledge base documents retrieved from FAISS vector database (Model: `all-MiniLM-L6-v2`)



IndustroSense AI

Responsible Multimodal Industrial Diagnostic System

Navigation

- Multimodal Diagnostics
- Maintenance Calculator
- Knowledge Base Explorer
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- Model Card & Governance

System Configuration

Retrieval Confidence Threshold

0.35

Deploy

Retrieved Technical Sources & Similarity Scores

Top-3 relevant knowledge base documents retrieved from FAISS vector database (Model: `all-MiniLM-L6-v2`)

DOC-001

73.4% Match

Electric Motor Maintenance Manual

Category: Electrical & Rotating Machinery
Equipment: Electric Motor

"Electric motors may overheat due to excessive electrical load, poor ventilation, damaged bearings, or inadequate lubrication. Regular inspection of the cooling ..."

DOC-006

56.4% Match

Industrial Incident Log

Category: Historical Incident Records
Equipment: Plant Operations & Motors

"A previous motor overheating incident was caused by worn bearings and insufficient lubrication. The issue was resolved by replacing the bearings and applying pr..."

DOC-002

43.9% Match

Electric Motor Bearing Guide

Category: Mechanical Components
Equipment: Electric Motor Bearings

"Damaged or poorly lubricated bearings can cause excessive friction, vibration, grinding sounds, and increased operating temperature. Bearings should be inspecte..."

Source & Similarity Transparency: Retrieval scores represent cosine similarity between semantic vector embeddings. Higher similarity indicates topical relevance in technical documentation, not diagnostic certainty.

Grounded Diagnostic Guidance



IndustroSense AI

Responsible Multimodal Industrial Diagnostic System

Navigation

- Multimodal Diagnostics
- Maintenance Calculator
- Knowledge Base Explorer
- Security & Audit Logs
- Model Card & Governance

System Configuration

Retrieval Confidence Threshold

0.35

Deploy

Grounded Diagnostic Guidance

Possible Causes

- Possible cause: Excessive electrical overload
- Possible cause: Poor ventilation / clogged cooling fan
- Possible cause: Damaged or worn bearings
- Possible cause: Inadequate or contaminated lubrication
- Possible cause: Extended operation past scheduled maintenance window

Recommended Actions

- Action: Measure operating current against nameplate full-load amps (FLA)
- Action: Inspect and clean cooling fins, fan cover, and ventilation paths
- Action: Check bearing temperature and perform acoustic/vibration testing
- Action: Replenish or replace lubrication grease according to manufacturer spec
- Action: Review past maintenance logs for overdue service intervals

View Complete Grounded Evidence Citations

Responsible AI & Industrial Safety Notice:

This system provides decision-support information and should not replace qualified engineers or technicians for safety-critical industrial decisions. Always follow Lockout/Tagout (LOTO) protocols and verify electrical/mechanical systems physically prior to servicing.